

Degree of Anthropogenic Land Disturbance Controls Fluvial Sediment Hysteresis

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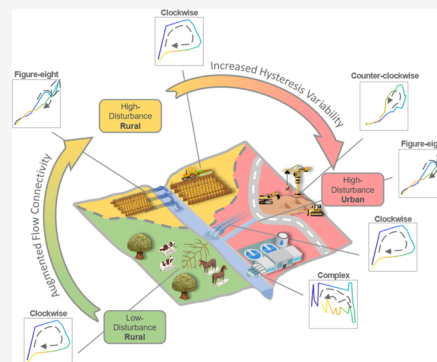
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ABSTRACT: Storm events dominate sediment delivery to stream corridors, but the effects of anthropogenic disturbances on altering the sources, pathways, and timing of delivery remain uncertain. To address this knowledge gap, we analyzed 849 events from over a decade of high-frequency turbidity data across five watersheds in an urbanization gradient. Sensing results suggested that hysteresis patterns evolved with land use from clockwise (low-rural) to figure-eight (high-rural) to counter-clockwise (high-urban), indicating a disturbance-driven shift of sediment provenance from proximal to distal. Sediment loading pathways in the lowest-disturbance rural watershed were dominated by a single hysteresis shape (>90% of export by clockwise events), whereas the most-disturbed urban basin had the greatest variability in loading pathways (~25% of export by clockwise, counter-clockwise, figure-eight, and complex events, respectively). Finally, wastewater treatment facilities modulated the release of “hungry-water” baseflow, causing more-rapid periods of high streamflow variance in catchments with a treatment facility (~4 h period) than in those without (~6 h period). Together, our results indicate that anthropogenic disturbances, including tile drainage, impervious surfaces, and roadway density, increase the connectivity of distally located sediment that would—in undisturbed basins—deposit along the sediment cascade. This information is important to watershed managers as they mitigate erosion in developing basins.

KEYWORDS: sediment transport, urbanization, hysteresis, spectral analysis, connectivity



1. INTRODUCTION

Sediment pollution of water bodies can result in aquatic habitat degradation,¹ hindered water supply,² and transport of pollutants;³ thus, it is of major concern to watershed managers. Effective sediment management necessitates insight into active sediment sources and pathways along with their response to hydroclimatic variables, geomorphological features, and land management operations. Though these parameters have been extensively researched in natural and rural basins,^{4–6} urban basins have received comparatively less study. While urban basins are recognized to result in increased sediment yields,⁷ greater bank erosion,^{8,9} and quicker discharge peaks,¹⁰ less is known about how anthropogenic disturbances affect the timing and sourcing of sediment in urbanizing landscapes. In this study, we define anthropogenic disturbances as network components that augment sediment connectivity and short-circuit disconnectivities along the sediment cascade.¹¹ Understanding the role of these disturbances is important as global modeling projections indicate land development will increase substantially to satisfy population growth and migration patterns.¹²

Anthropogenic disturbances to sediment connectivity can influence hydrological responses by adding new pathways to streams, including by way of tile-drains, storm sewers, and drainage ditches.¹⁰ In highly disturbed urban settings, storm

sewers and drainage ditches are built to convey runoff from impervious surfaces, such as roadways, and thus increase the potential for sediment connectivity through flow channelization and increased conveyance.^{7,9} On the other hand, in disturbed agricultural settings, tile drainage can route sediment from distal upland locations directly to stream networks, thus circumventing vegetation, flow limitation, and other disconnectivities.¹³ Despite recognition of the importance of these disturbances, there is a lack of quantitative assessment of their influence on the timing and magnitude of sediment delivery.

Determining the sources and pathways of sediment is challenging because transport dynamics are nonlinear and most delivery occurs over short event timescales.^{14,15} In this regard, high-frequency sensing of turbidity coupled with hysteresis analysis helps resolve these challenges.^{16,17} Hysteresis is defined as the asynchronous relationship between suspended sediment concentration and event discharge, and it can be used

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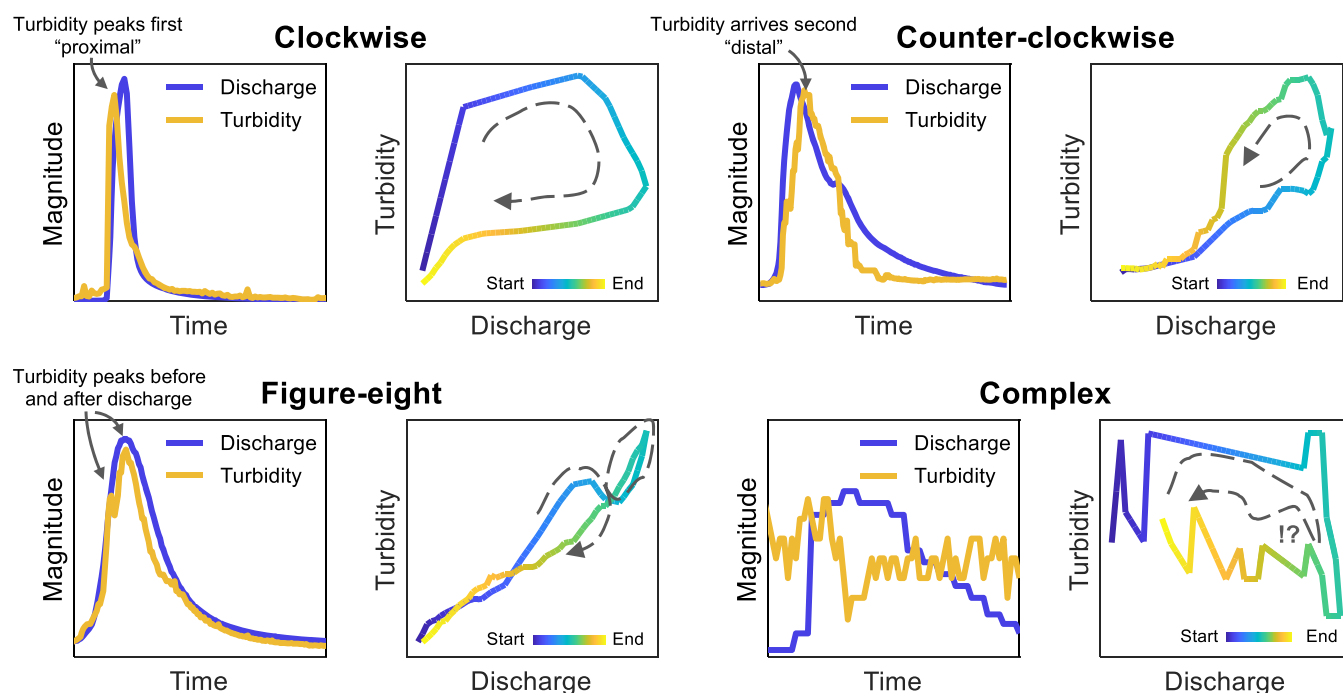


Figure 1. Conceptual sediment hysteresis archetypes, including clockwise, counter-clockwise, figure-eight, and complex hysteresis. For each shape, the left panel plots show the storm hydrograph and sedigraph, whereas the right panel plots show hysteresis behavior. The complex event shows unexpected dilution (?) for sediment.

to infer the source of sediment relative to the stream.^{18,19} There exist four primary sediment hysteresis archetypes: clockwise, counter-clockwise, figure-eight, and complex (Figure 1). For clockwise events, the turbidity signal peaks before discharge, indicating proximal sediment sourcing, whereas the opposite is true for counter-clockwise events, indicating distal sediment sourcing. On the other hand, figure-eight patterns arise from multiple turbidity peaks and occur due to a combination of proximal and distal sourcing. Finally, during complex hysteresis, system dynamics are not easily inferred and phenomena such as sediment dilution during an event may be observed. Because turbidity and flow are measured at a single point in space, observed hysteresis represents an imprint of many upstream structural properties, including drainage area, land use, topography, and pedology.¹⁴ For the purposes of isolating the impacts of anthropogenic disturbances on sediment hysteresis, it is vital to select study locations that differ in land use but are similar with respect to other basin properties.

Dynamic hydroclimatological characteristics, such as storm intensity and duration, work in conjunction with structural watershed properties to influence observed stream hysteresis.²⁰ To assess the strength of linkages between event characteristics and stream response, researchers use tools like principal component analysis (PCA)²¹ and significance testing.²² PCA describes correlations between explanatory variables and reorganizes these variables into a few linear components, which represent underlying similarities in correlated variables.²¹ We explore data associations to identify which hydroclimatological variables are significantly linked with a particular hysteresis response (e.g., smaller-than-average values of storm duration tied to counter-clockwise hysteresis).²² To complement PCA and data exploration, spectral analysis can inform whether streamflow and turbidity time series follow temporal scaling laws (e.g., long-term autocorrelation or

randomness) by transforming the signals to the frequency domain.²³ This powerful transformation allows for decomposing the variance in a time series to its corresponding frequencies.²⁴ As hysteresis is inherently linked to the asynchronous temporal behavior of sediment and water, spectral analysis can identify dominant frequencies involved in recurring anthropogenic phenomena, such as periodic wastewater discharge, that may exert control on the timing of turbidity pulse arrival.²⁵ Despite the widespread use of PCA, significance testing, and spectral analysis, their application to investigating how anthropogenic disturbances affect the sourcing, timing, and frequencies controlling sediment transport is unrealized.

The overall objective of this study was to estimate how anthropogenic disturbances, such as roadway and tile drainage, impact the sourcing and pathways of sediment transport. We hypothesized that watersheds with greater anthropogenic disturbance will have greater clockwise, proximal hysteresis as delivery of flashy, sediment-free runoff preferentially erodes near-stream sources. We tested this hypothesis by evaluating high-frequency turbidity sensor data for 849 storm events, over the course of a decade, across five watersheds in an urbanizing landscape. We performed hysteresis analysis to characterize sediment sourcing and timing, PCA and data significance testing to investigate the controls on hysteresis, and spectral analysis to assess the temporal scales driving sediment transport. Finally, we provide a conceptual model of how the type and degree of anthropogenic land disturbance alter sediment pathways, connectivity, and transport mechanisms.

2. STUDY SITE

To meet the objectives of this study, five watersheds in rapidly urbanizing Johnson County, Kansas, were selected as study sites (Figure 2). Land use in Johnson County varies from rural in the southwestern areas to completely urban in the

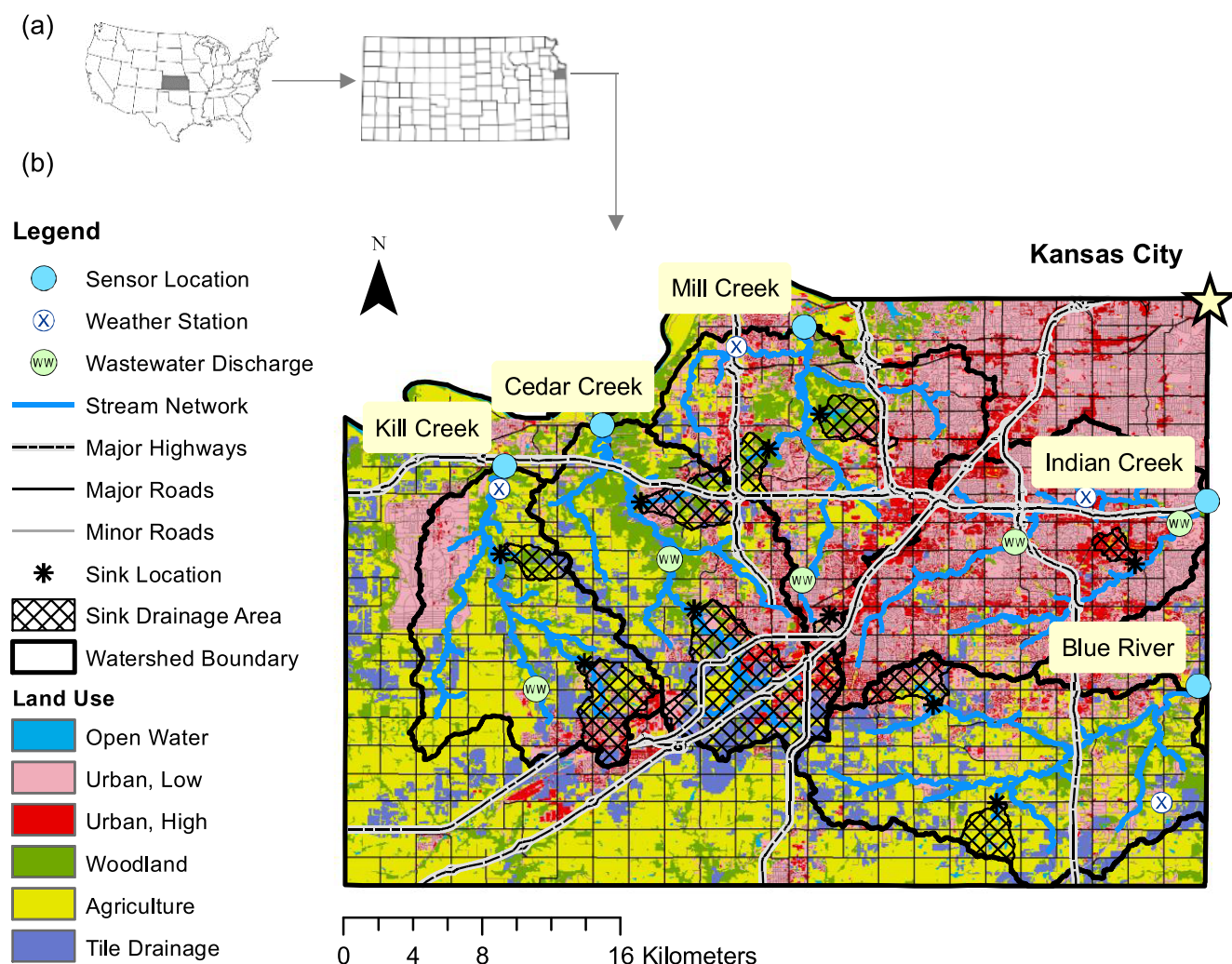


Figure 2. (a) Location of Johnson County, Kansas. (b) Sensor locations, weather stations, wastewater treatment facility discharges, channel networks, major and minor roadways, sediment sinks (e.g., lakes and reservoirs), land use, and tile-drained area of study watersheds.

Table 1. Drainage Area, Percent Watershed Draining into Sediment Sinks, Percent Nonsink Area Drained by Tiles, Roadway Density, Mean Stream Discharge, Wastewater Treatment Facility (WWTF) Permitted Discharge, and Land Use Distribution for Study Watersheds

watershed	area (km ²)	sink area (%)	tiled area (%)	road density (km/km ²)	mean flow (m ³ /s)	WWTF permit (m ³ /s)	land use (%)			
							urban	rural	forest	water
Kill Creek	130	13.1	14.8	2.76	0.92	0.11	26.4	55.4	17.2	1.0
Cedar Creek	153	35.3	12.2	3.34	1.32	0.13	30.5	43.9	24.3	1.3
Blue River	170	7.6	18.6	2.69	1.68	0.00	21.1	64.1	13.6	1.2
Mill Creek	150	10.0	5.5	6.25	1.56	0.14	61.8	23.4	14.2	0.6
Indian Creek	168	1.2	3.0	9.33	2.70	0.96	89.4	6.8	3.2	0.2

metropolitan parts of Kansas City to the northeast. Johnson County is relatively flat (mean slope of 3.7°) and has a temperate climate with mean annual temperature and precipitation of 13.7 ± 0.4 °C and 958 ± 211 mm per year, respectively. The studied watersheds fall along an urban-to-rural transition, and we broadly classify them into three categories: low-disturbance rural (Kill Creek and Cedar Creek), high-disturbance rural (Blue River), and high-disturbance urban (Mill Creek and Indian Creek). Urban land use is least in Blue River (21%) and greatest in Indian Creek (89%) and is generally proportional to the density of

roadways in a basin (Table 1). The style of urban development is primarily residential (urban, low; Figure 1) with moderate commercial and industrial zones (urban, high; Figure 1). Blue River has few impervious surfaces (4%), whereas Indian Creek has many (30%). Sediment sinks (e.g., lakes and reservoirs) exist throughout the county, most prominently in Cedar Creek where they drain 35.3% of basin area. Cropland tile drainage varies from negligible (3.0% of Indian Creek) to prominent (18.6% of Blue River). All watersheds, except for Blue River, have wastewater treatment plant facility releases into their main stems, ranging in maximum permitted discharges of 0.1–

1.0 m³/s (Table 1). Despite the watersheds' similar drainage areas, slopes, climates, and soils, the distinct differences in land use offer an opportunity for aquatic sensing to elucidate the impacts of anthropogenic disturbances on altering sediment transport.

3. MATERIALS AND METHODS

To investigate fluvial hysteresis, 15 min high-frequency turbidity and discharge data were obtained from the United States Geological Survey (Figure S1), as part of their extensive water quality monitoring program in Johnson County.²⁶ Turbidity and discharge data were monitored at watershed pourpoints using YSI model 6136 multiparameter sensors and piezometers, respectively. The shortest data-record was at Kill Creek, lasting 4 years, whereas the longest data-record was at Indian Creek, lasting 10 years (Figure S1). Eighteen to 30 suspended sediment samples were collected at each pour point and regressions between turbidity and suspended sediment concentration (SSC) were very strong ranging from R^2 of 0.80 at Mill Creek to 0.93 at Kill Creek (Figure S2). Short data gaps (<3 h) resulting from sensor malfunction or routine maintenance were filled by the last previous value, whereas longer gaps were left in place. Road networks were extracted from the topologically integrated geographic encoding and referencing (TIGER) data set.²⁷ Tile-drained area was estimated from AgTile-US, a predictive 30 m geospatial model with 86% field-truth accuracy.²⁸ Finally, precipitation data were downloaded from the nearest weather station to each pour point (the furthest being 5 km from a sensor site) using the Johnson County StormWatch database.²⁹

3.1. High-Frequency Sensing of Sediment Hysteresis.

Hysteresis was analyzed for all storm events during the study period to understand the timing, sources, and pathways of sediment. We defined a storm event as a period beginning with a rising limb that is at least three times initial baseflow and concluding with either a return to baseflow or the beginning of another event. For each event, we generated time series and hysteresis plots (as in Figure 1), and visually classified event hysteresis as clockwise, counter-clockwise, figure-eight, and complex. Sediment yield was calculated for each event by integrating the sediment load ($Q \times SSC$) over the duration of the event. Thereafter, we calculated the proportion of events classified by each hysteresis type and the proportion of sediment yield contributed by each hysteresis type.³⁰ To assess uncertainty in sediment yield estimates, we generate prediction bounds for stage-discharge and turbidity-SSC time series, model 100 iterations of error time series for both signals, and randomly sample 10 000 iterations of the two signals to generate statistics.³¹ Finally, we investigated seasonality of hysteresis for wet (April to September) and dry (October to March) seasons.

We utilized the hysteresis index (HI) and flushing index (FI) to semi-quantitatively characterize the direction and magnitude, respectively, of hysteresis.^{31–34} To compute HI, stream-flow and turbidity data were normalized for each event and the hydrograph was split into 5% flow intervals. Each flow interval has a rising and falling limb component, where the corresponding turbidity values are retrieved. From there, the HI was calculated as the difference between normalized turbidity on the rising (X_i^{RL}) and falling (X_i^{FL}) limb at a flow interval (i)

$$HI_i = X_i^{RL} - X_i^{FL} \quad (1)$$

The mean of all calculated HI_i values for an event was then considered as an overall HI event value.¹⁸ Positive HI values suggest a clockwise hysteresis pattern, negative HI values indicate a counter-clockwise direction, and the absolute value of HI shows the magnitude of hysteresis. To quantify the intensity of sediment mobilization, FI was calculated by taking the difference between the normalized turbidity value at peak discharge (X_{Q-peak}) and at the beginning of the storm ($X_{initial}$)

$$FI = X_{Q-peak} - X_{initial} \quad (2)$$

FI ranges from -1 to $+1$, where negative values and positive values indicate dilution and flushing, respectively, during the rising limb of an event.³⁴

3.2. Statistical Analysis of Sediment Hysteresis.

3.2.1. Principal Components of Sediment Drivers. Principal component analysis was performed to generally describe sediment drivers for each of the four hysteresis categories. We started with eleven potential drivers (i.e., variables; Table S1) and used PCA to reduce the dimensionality of the variable space while maintaining as much of the original information as possible. The original variables were grouped by similarity into three categories: runoff, antecedent, and rainfall. Runoff variables include the storm duration (t_{storm}), time to peak flow (t_{peak}), peak storm flow (Q_{peak}), and flood intensity (I_{flood}). Antecedent variables include the 3-day (AP_3) and 14-day (AP_{14}) antecedent precipitation amounts, initial baseflow (Q_{base}), the time gap between the previous and current storm peaks (t_{gap}). Rainfall variables include the total storm precipitation (P_{storm}), the maximum 30-min rainfall intensity (I_{30}), and the duration of rainfall (t_{rain}). To understand how the explanatory variables help influence hysteresis shape, we selected flow-weighted sediment concentration (SSC_{fw}) as our response variable. SSC_{fw} is defined as the total event suspended sediment load divided by the total event flow volume and has advantage over other response variables as it eliminates the effects of hydrological variability of storms.²¹ Potential controls were then determined by which of the eleven explanatory variables occupied the same dimensionally reduced space as the response variable (SSC_{fw}). PCA was calculated in the JMP statistical discovery software.³⁵ Finally, we investigated direct correlations between explanatory variables and the target variable by calculating the Pearson correlation coefficient (r), using an r value greater than 0.50 to indicate strong correlation.⁵

3.2.2. Exploring Data Associations of Hysteresis Controls.

We explored data associations of the explanatory variables with the aim of gaining a better understanding of how each variable may exert control on fluvial hysteresis. First, the eleven explanatory variables were calculated for all classified storm events. Thereafter, these bulk distributions were subdivided into four bins, one for each hysteresis shape (i.e., clockwise, counter-clockwise, figure-eight, and complex). A Mann–Whitney significance test was then conducted, comparing the variable distribution from particular hysteresis events (e.g., clockwise) against the variable distribution from all other hysteresis events (e.g., counter-clockwise, figure-eight, and complex). Variables that were significantly different for a given hysteresis type compared to all other events were assumed to be informative to explaining processes contributing to that hysteresis pattern. We tested significance at three levels, $\alpha = 0.10$, 0.05, and 0.01, as in other studies.²² We utilize a broad range of significance levels as the scale of our study (~ 150 km² watersheds draining to a single point under the co-action of

geologic, climatic, and anthropogenic drivers) makes it difficult to entirely separate the influences of different variables, as may be the case in smaller, more-controlled settings. Finally, we estimated the effect size of significant relationships using the nonparametric Cliff's δ statistic and benchmarked δ against the criteria of 0.11 = small effect, 0.28 = medium effect, and 0.43 = large effect.^{36,37} The reader is referred to Table S1 for a further description of how variables were calculated and how they influence hysteresis shape.

3.2.3. Spectral Analysis of Sediment Timing. Fourier spectral analysis was used to investigate dominant frequencies and self-similarity of streamflow and turbidity signals.²³ The Fourier transform decomposes a time series signal into harmonic functions, each with its own frequency and amplitude, which can elucidate seasonal patterns and anthropogenic influences, such as wastewater discharge. To mathematically investigate time series signals in the frequency domain, we calculate the power spectral density (PSD) of a time series using Welch's algorithm with a Hanning window of 3000 data points and a low-pass filter with three-point smoothing.³⁸ If a time fractal process exists in the data, it will follow a power law relationship between the spectral power (S) and frequency (f) with a temporal scaling of $S(f) \propto f^\beta$, where β is the scaling exponent that represents signal persistence (loosely, the "memory"). Signals with β less than 1 are considered to be dominated by Gaussian noise, β values between 1 and 3 are considered fractional Brownian movement with some persistence, and β values larger than 3 show strong autocorrelation.^{39,40} To gain a better insight into the magnitude of variations explained by each frequency, the spectra can be shown in a variance-preserving form. The variance-preserving form, also called the premultiplied spectra, is a semi-logarithmic plot in which the x -axis (logarithmic scale) remains as frequency but the y -axis (linear scale) is presented as the product of the S by its corresponding f .³⁸ Thereafter, we normalized each watershed's variance-preserving spectra by their respective peak values to allow for direct comparison of the frequency at which most of the variance in the signal occurs, which may be indicative of human influences on hydrology.

4. RESULTS

4.1. High-Frequency Sensing of Sediment Hysteresis.

Between 2002 and 2014, a total of 849 distinct stream events were identified across all watersheds, ranging from 66 in Kill Creek to 293 in Indian Creek (Figure S1 and Table S2). The number of events were broadly consistent with the length of data availability, but urban watersheds tended to have more events per year than agricultural watersheds as urban basins required a lower rainfall threshold to generate a stream response (Table S3). Kill Creek and Cedar Creek, the least disturbed agricultural watersheds, showed nearly identical hysteresis distributions with approximately 34% clockwise, 35% counter-clockwise, and 20% figure-eight patterns. Blue River, the most-disturbed agricultural watershed as evidenced by greater crop cover and tile drainage, differed from the other agricultural watersheds with 15% clockwise, 36% counter-clockwise, and 41% figure-eight hysteresis. The urban watersheds, Mill Creek and Indian Creek, had distinct behavior from agricultural watersheds with approximately 8% clockwise, 60% counter-clockwise, and 21% figure-eight. All watersheds have approximately 10% complex or uncharacterizable events. In general, results indicate that low-disturbed agricultural water-

sheds have a high proportion of clockwise events and, as rural disturbances increase, the watershed hysteresis response transitions to figure-eight. Finally, as anthropogenic disturbances transition from rural to urban, figure-eight patterns give way to counter-clockwise hysteresis.

The average HI was negative for all watersheds, indicating that discharge peaks sediment (i.e., counter-clockwise hysteresis) for most events (Table S2). The absolute value of HI increased with the intensity of disturbance from -0.01 for lowest-disturbed rural to -0.23 for highest-disturbed urban watersheds. These results indicate that greater asynchronous behavior (i.e., lag time) occurs between discharge and turbidity in urban watersheds compared to rural basins. Further, FI values showed strong mobilization of sediment on the rising limb of storm events for all watersheds but were not necessarily related to land use or extent of roadways, tile drainage, or impervious area.

Sediment flux from the watersheds broadly followed the degree of anthropogenic disturbance (Figure S3), ranging from 0.096 ± 0.002 to 0.344 ± 0.005 kt km⁻² yr⁻¹ in Kill Creek and Indian Creek, respectively. However, the regularity by which a hysteresis type occurs does not reflect the amount of sediment yielded by that type (Figure 3). In all watersheds, clockwise events transported a greater fraction of sediment than the frequency by which they occurred. This is most well demonstrated in Kill Creek where clockwise hysteresis constituted 35% of events but 92% of the sediment load. On the other hand, sediment loads from counter-clockwise hysteresis were always smaller than their frequency of occurrence. For example, counter-clockwise events occur 65% of the time in Mill Creek but export only 36% of the sediment. Seasonal patterns could also be observed in the frequency of hysteresis occurrence (Figure S4). For all watersheds, clockwise events gained prominence in the wet season while counter-clockwise events gained prominence in the dry season. On average, clockwise events are $1.54 (\pm 0.37)$ times more likely to occur in the wet season and counter-clockwise events are $1.52 (\pm 0.50)$ times more likely to occur in the dry season.

4.2. Statistical Analysis of Sediment Hysteresis.

4.2.1. Principal Components of Sediment Drivers. Across all hysteresis types and basins, the variance explained by the first two principal components ranged from 46.8 to 78.8% (Figure 4). Based on grouping of explanatory variables, the first principal component (PC 1) typically represents storm magnitude and intensity (Q_{peak} , I_{flood} , P_{storm} , I_{30}), whereas the second principal component (PC 2) indicates antecedent conditions (Q_{base} , AP_3 , AP_{14}). Thus, in most PCA plots, the target variable (suspended sediment concentration, SSC_{fw}) was primarily associated with storm intensity with secondary association to antecedent wetness. Interestingly, this was not the case for clockwise (proximal) hysteresis in Cedar Creek where antecedent conditions were more important. Thirty-five percent of Cedar Creek's basin area drains to sediment sinks (e.g., lakes and reservoirs, Table 1); thus, the initial intensity of storms is attenuated by network disconnectivities and the role of baseflow and watershed wetness becomes more prominent. The highly tile-drained basin, Blue River (18.6% tile-drained, Table 1), had interesting behavior with respect to counter-clockwise (distal) hysteresis. Large rainfall amounts and flood intensities were of lesser import for generating high sediment concentrations (SSC_{fw}) in Blue River, potentially because wetting of the soil could lead to gradual piping of distal

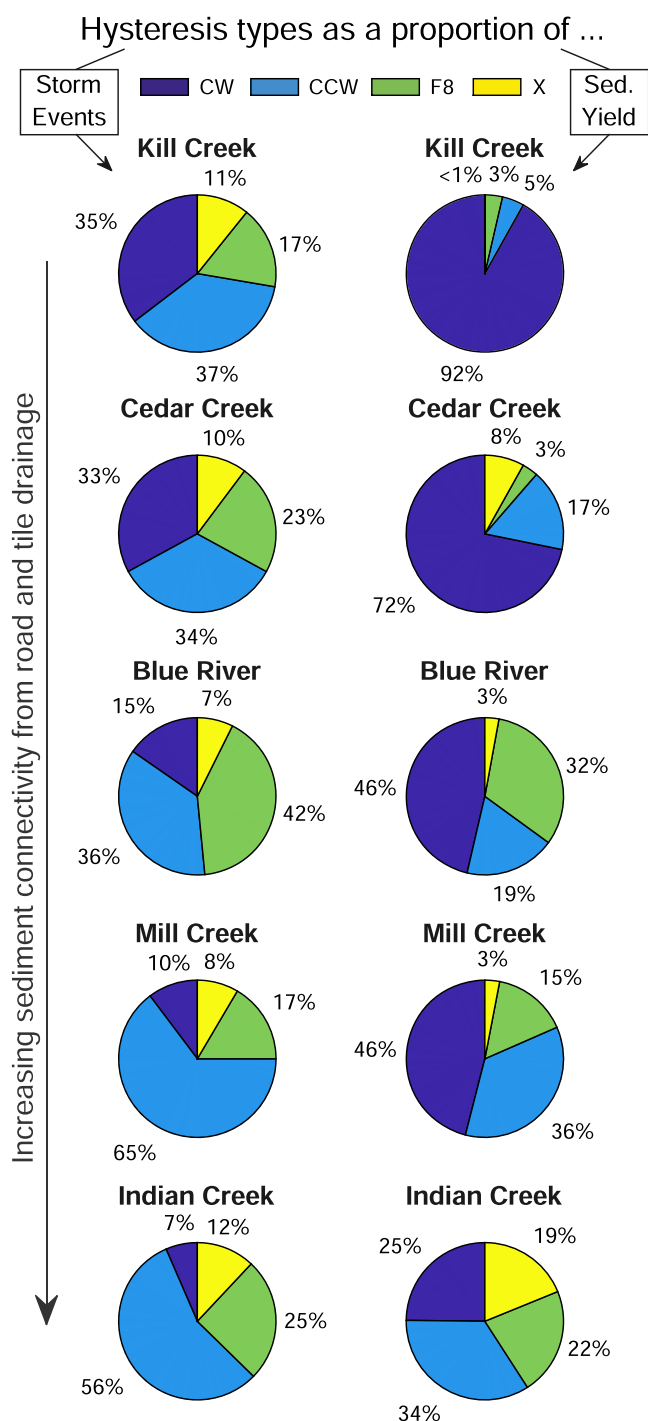


Figure 3. Hysteresis types for each watershed as a proportion of storm-event classification (left column) and contribution to total sediment yield (right column) for our five watersheds. Legend: CW (clockwise), CCW (counter-clockwise), F8 (figure-eight), and X (complex).

sediment through the basin's extensive tile network. In figure-eight hysteresis, where proximal and distal sediment sourcing is prominent, baseflow (Q_{base}) was an influential driver of SSC_{fw} in all basins as it provides the conditions for a proximal, prewetted sediment pulse followed by the connectivity and delivery of storm-activated, distal sediment. This was most true in the watersheds which have the least disturbance, Kill Creek and Cedar Creek (strong correlation $r > 0.5$ of Q_{base} with

SSC_{fw}). We inferred this observation to mean that prewetted conditions aid in "priming" connectivity in these lesser-connected basins, whereas the highly disturbed basins do not require equal priming as their connectivity is already augmented by anthropogenic activities like tile and roadway drainage. As expected, additional drivers beyond storm intensity are difficult to ascertain in complex hysteresis, where all events not fitting clockwise, counter-clockwise, and figure-eight patterns were grouped.

4.2.2. Exploring Data Associations of Hysteresis Controls.

The results of significance testing to explore which variables were more conducive to producing certain hysteresis types indicated great variability in controls across watersheds (Table 2). For runoff variables, complex events typically occurred when a storm event had smaller than mean peak discharge (Q_{peak}) and flood intensity (I_{flood}). On the other hand, larger than average Q_{peak} and I_{flood} caused clockwise events in rural basins but figure-eight patterns in the most-disturbed basin (Indian Creek). Regarding antecedent conditions, high baseflows (Q_{base}) were correlated with clockwise events in the lesser-disturbed rural watersheds but counter-clockwise events in the most-disturbed rural (Blue River) and urban (Indian Creek) watersheds. Further, shorter times between storm-event peaks (t_{gap}) resulted in clockwise events in the agricultural watersheds. Finally, rainfall variables had the most consistent effects across most watersheds. Small storm precipitation amounts and 30-min intensities (P_{storm} and I_{30}) were associated with counter-clockwise behavior, whereas large P_{storm} and I_{30} caused clockwise behavior in nearly every watershed. Effect sizes, as calculated using Cliff's δ , were typically larger in the lesser-disturbed watersheds potentially because of the strong imprint of a single hysteresis shape (i.e., clockwise) in controlling overall sediment export from these watersheds (see "Kill Creek" and "Cedar Creek" in Figure 3).

4.2.3. Spectral Analysis of Sediment Timing. In all watersheds, we observed power law behavior in streamflow, turbidity, sediment concentration, and sediment load time series (Table S4), suggesting "memory" in the signals. The scaling exponent (β) for streamflow ranged from -3.70 to -4.56 , indicating a high degree of autocorrelation.⁴⁰ The weakest streamflow autocorrelation occurred in Cedar Creek ($\beta = -3.70$) and could be due to the extensive of sediment sinks in the basin, i.e., 35% of basin area drains to lakes and reservoirs (Table 1), which interfere with autocorrelation as the impact of sinks can be nonperiodic. The strongest streamflow autocorrelation was observed in Kill Creek ($\beta = -4.56$), the least disturbed watershed in our study with a consistent $0.11 \text{ m}^3/\text{s}$ wastewater treatment plant discharge. The β values for turbidity (-1.96 to -2.58 , Figure 5a) lie in the nonstationary fractal Brownian motion zone (between -1 and -3), which implies some degree of autocorrelation.²³ We could not identify a correlation between the degree of anthropogenic disturbance and signal autocorrelation, suggesting that factors other than land use control water and sediment autocorrelation. However, the periodicity of streamflow and sediment signals in each watershed can be informative to understanding the physical processes that create asynchronous behavior between flow and turbidity, i.e., hysteresis.

The variance-preserving spectral plots show the distribution of variance-containing frequencies in the streamflow signature (Figure 5b). The location of peak frequency varied from 3.8 per day in Blue River to 7.4 per day in Indian Creek, corresponding to peak variance-contributing periods of 6.3 and

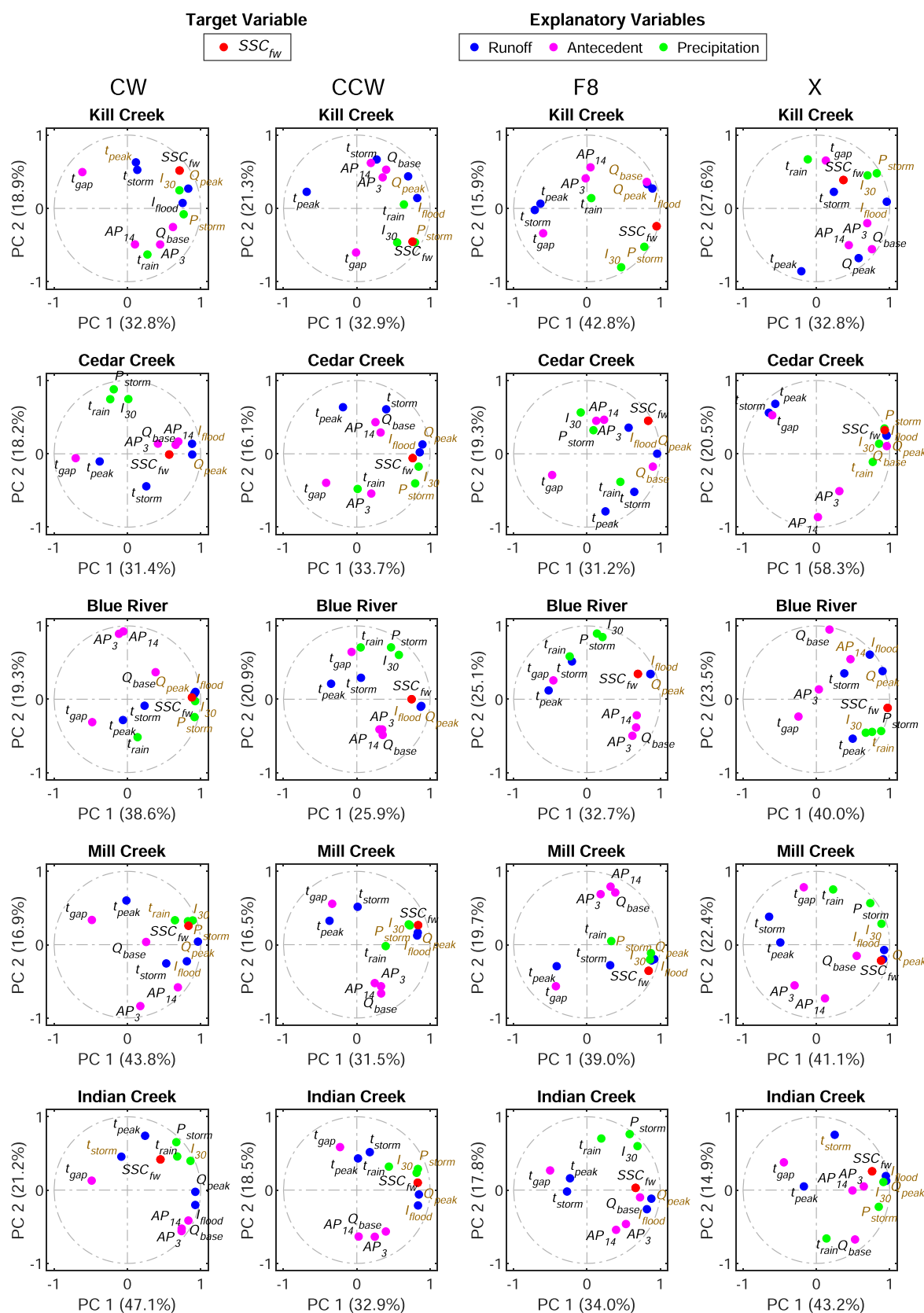


Figure 4. Principal component analysis for clockwise (CW), counter-clockwise (CCW), figure-eight (F8), and complex (X) hysteresis patterns as driven by runoff, antecedent, and precipitation controls. Explanatory variables with a strong correlation ($r > 0.5$) with the target variable, SSC_{fw} are labeled with gold font. Variables with strong association group together along PCs. See Table S1 for a full definition of all variables. Note: some variables may have further correlation with SSC_{fw} beyond the first two principal components (e.g., PC 3, PC 4, etc.).

Table 2. Impact of Runoff, Antecedent, and Rainfall Variables on Determining Hysteresis Shape (Clockwise $\cup$, Counter-clockwise $\cap$, Figure-Eight ∞ , and Complex $\times$) for the Five Studied Watersheds^a

Variable Lower Than Mean						Variable Greater Than Mean					
← increasing sediment connectivity						increasing sediment connectivity →					
	Indian Creek	Mill Creek	Blue River	Cedar Creek	Kill Creek	Variable	Kill Creek	Cedar Creek	Blue River	Mill Creek	Indian Creek
Runoff	∪ *		× ∘			t_{storm}			∞		
		× *	× *		× **	t_{peak}		× ∘			∪ *
			× *		× **	Q_{peak}	∪ **	∪ *		∪	∞ *
	∪ *		× *		× **	I_{flood}	∪ **	∪			∞ *
Antecedent					∞ **	AP_3	∪ **	∪			
					× ∘	AP_{14}	∪ **				
	∞		∪, × *		∪, × **	Q_{base}	∪ **	∪ *	∪ **		∪ *
				∪	∪ *	t_{gap}	∪	∞ *	× *		
Rainfall	∪ **	∪ **	∪ *	∞ ∘	∪ **, × *	P_{storm}	∪ **	∪ **		∪ **	∪ *, ∞ *
	∪ **	∪ **	∪ *	∞	∪ *	I_{30}	∪ **			∪ **	∪, ∞ **
				∞ *	∪ *	t_{rain}	∪ **	∪ **			

^aSignificant differences at the 0.10, 0.05, and 0.01 levels are indicated by no superscript, a single asterisk (*), and double asterisks (**), respectively. Small, medium, and large effect sizes are indicated by no subscript, a single ring ($\circ$), and double rings (∞), respectively. An example of interpreting the table: the value of peak flow (Q_{peak}) was significantly higher ($p < 0.01$) and had a large effect size for events causing clockwise ($\cup$) hysteresis in Kill Creek compared to Q_{peak} for all other hysteresis types. For a full explanation of variable definitions and impacts to hysteresis, see Table S1.

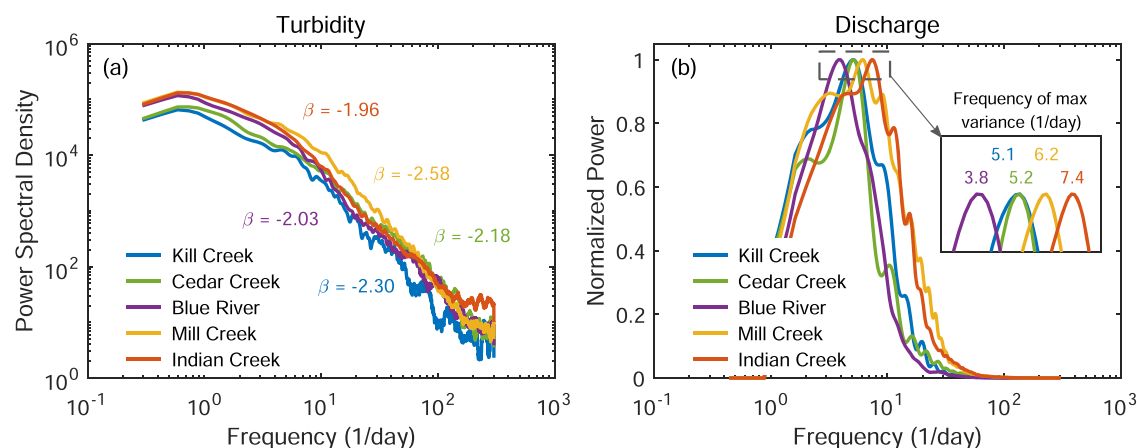


Figure 5. (a) Power spectral density diagram of turbidity data with spectral slopes (β) listed. (b) Normalized variance-preserving spectral plot for streamflow with an inset of the frequency of maximum variance in the signal.

3.2 h, respectively. In general, the agricultural watersheds (Kill Creek, Cedar Creek, and Blue River) had a much lower period (5.2 ± 0.9 h) of maximum variance contribution compared to the urban watersheds (Mill Creek and Indian Creek; 3.6 ± 0.4 h). Further, there are several local maxima peaks in frequency ranges higher than the global maxima, indicating other important frequencies that contribute to streamflow variance. For example, in the most-urban watershed, Indian Creek, there are two additional local maxima frequencies at 11.3 per day (2.1 h) and 15.3 per day (1.6 h). Kill Creek, Cedar Creek, and Mill Creek, watersheds that receive ~ 0.1 m³/s of wastewater treatment facility discharge from 10 km upstream, all had a secondary variance maximum at 9.5 per day, corresponding to a period of 2.5 h. Blue River, the only watershed without a wastewater treatment facility, has no secondary variance maxima above its peak power.

5. DISCUSSION

We used hysteresis modeling along with statistical and spectral tools to gain more insight into coupled land use and hydroclimatological effects on sediment sources and transport pathways. We posed the hypothesis that watersheds with greater anthropogenic disturbance will have greater clockwise, proximal hysteresis as delivery of flashy, sediment-free runoff preferentially erodes near-stream sources. Our results lead us to reject this hypothesis. Anthropogenic disturbances, including roadway and tile drainage, increase the connectivity of distal sediment zones, circumvent disconnectivities that would otherwise cause sedimentation, and result in counter-clockwise fluvial hysteresis (Figure 6).

5.1. Sourcing and Pathways of Sediment in an Urbanizing Landscape. Low-disturbance rural watersheds (Kill Creek and Cedar Creek) had the greatest incidence of

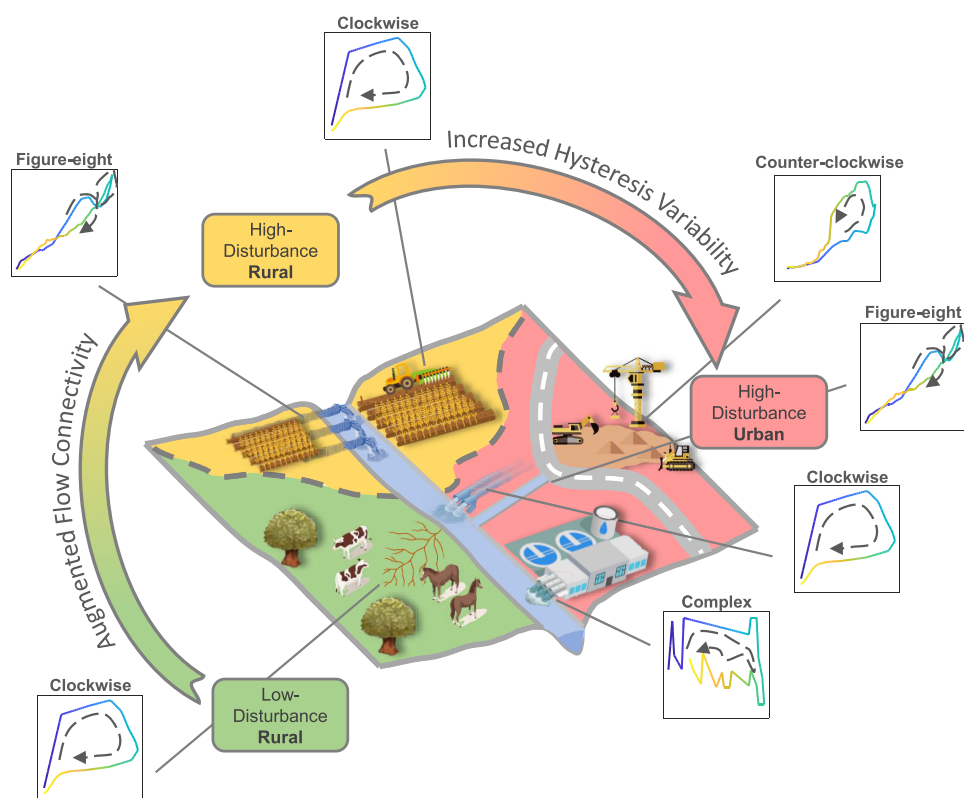


Figure 6. Conceptual model of how anthropogenic disturbance and land use change control the timing, pathways, and hysteresis of basin-scale sediment transport. The x - and y -axes on each hysteresis plot represent discharge and turbidity, respectively.

clockwise hysteresis of all watersheds, indicating proximal sediment contribution (Figure 6). This proximal sediment mobilization was driven by large flood intensities and long rainfall durations. Taken together, these drivers imply that low-disturbance agricultural watersheds take considerable rainfall to activate their near-channel sediment sources. However, near-channel sourcing does not necessarily imply subsurface (i.e., streambank) sediment erosion as there could be proximal agricultural fields and pastures that require significant hydrologic activity to overcome disconnectivities along the field-to-streamflow path.^{11,41} Further, historic upland sediment deposits in floodplains and streambeds could be resuspended, providing an ostensibly “near-channel” source that was originally very distal to the stream.⁴²

The high-disturbance rural watershed (Blue River) showed a predominance of figure-eight hysteresis, indicating that both proximal and distal sediment sources contribute in significant quantities to the stream (Figure 6). The mixed proximal-distal mobilization of sediment was primarily driven by long storm-event duration, allowing for sufficient wetting of watershed transport pathways. This behavior is attributed to the higher connectivity of upland sediment to streams due to the presence of connectivity-augmenting features such as artificial tile drainage networks.¹³ At the field scale, subsurface drainage has been shown to pump surficial soil sediment directly to stream networks via macropore and preferential flow pathway connection to tiles.⁴³ This high-disturbance hypothesis is supported by Blue River having (1) the greatest rural land use, much of which is cropland, and (2) the greatest density of tile drainage of all watersheds (Table 1). In a sense, figure-eight hysteresis behavior is a transitional stage of watershed sediment transport dynamics, situated between clockwise and

counter-clockwise behavior, and includes low-connectivity features of vegetated watersheds together with augmented-connectivity of anthropogenically drained landscapes.⁴⁴

In the high-disturbance urban watersheds (Mill Creek and Indian Creek), we observed the greatest occurrence of counter-clockwise hysteresis, suggesting the significance of distal sediment sourcing (Figure 6). In the most-disturbed urban basin (Indian Creek), this distal behavior was primarily driven by large baseflow and long time to peak flow. In both urban basins, lower-than-average rainfall amounts and intensities also resulted in distal sourcing. Together, these drivers imply that the urban watersheds are highly reactive to all precipitation events and that there exists a large, highly transportable sediment supply, which takes some time to arrive to the outlet. Again, it is important to note that “distal” sourcing does not necessarily imply upland hillslope sediment provenance, rather we hypothesize that the distal sourcing is really bank material connected from distal veins of the dendritic stream network. Recent radioisotope tracing of sediment in Johnson County supports this idea as bank sediment constitutes, on average, 59% (Kill Creek) to 73% (Indian Creek) of the fluvial sediment load.⁴⁵ This bank material is mobilized using runoff from impervious surfaces and construction sites^{46–48} and is then routed to the mainstream corridor by way of dense urban roadway drainage (2–3 times greater roadway density than in rural basins, Table 1), where sediment connectivity remains elevated.^{44,49}

Wastewater treatment facility discharge, which is greatest in the studied urban watersheds, provides a constant source of water to keep material entrained and sediment transport capacity high.^{50,51} For example, in the most-urban Indian Creek, where plant discharges are 7 times greater than in any

of the other watersheds (Table 1), permitted discharge constitutes up to an average of 80% of streamflow (Figure S5), providing significant energy for transport. Another potential sediment source could be sewer overflows, which have been shown to contribute substantial fine-grained sediment to urban corridors.⁴⁸ This argument is supported by the larger contribution of complex, difficult-to-characterize hysteresis patterns in Indian Creek (Figure 3). To mitigate the incidence of sewer overflow events, the Kansas City metro area is undergoing a \$2.5 billion renovation of its sewer system. Thus, watershed managers should consider not only the land use and likely sources of sediment but also the most likely transport mechanisms, which can be informed by hysteresis analysis.

5.2. Anthropogenic Disturbance Augments Connectivity of Sediment. The degree and type of anthropogenic disturbance augment connectivity and add new storm-event controls on basin-scale sediment transport (Figure 6). Sediment loading from low-disturbance agricultural watersheds, such as Kill Creek and Cedar Creek, is primarily dominated by a single hysteresis type: clockwise (Figure 3). Clockwise events constitute 82% of sediment loading from low-disturbance rural watersheds despite making up only one-third of events. On the other hand, as disturbance increases, the variability in sediment export mechanisms also increases. In Blue River, the high-disturbance agriculture watershed, sediment export is split between two hysteresis types (46% by clockwise and 32% by figure-eight) rather than just a single shape. This variability in transport mechanisms increases even further in urban watersheds. In the most-disturbed urban watershed, Indian Creek, all four hysteresis types (clockwise, counter-clockwise, figure-eight, and complex) contribute considerably to the overall sediment load (25, 34, 22, and 19%, respectively). This variability in transport contrasts somewhat with the increasing magnitude of mean HI with increased disturbance (Table S2), which would suggest the shift toward a single, dominant hysteresis pattern. However, this contrast simply reinforces the need to investigate not only the proportion of events categorized by hysteresis shape but also the load carried by each shape.⁴¹ The augmentation of marginal hysteresis types serves as evidence of increased variability of sediment pathways, such as contributions from roadway drainage ditches, artificial drainage tiles, and impervious channels. Additionally, fragmented urban landscapes may exhibit smaller-scale sediment supply zones that, when mobilized, exhibit higher variability in their timing of arrival.

In the most anthropogenically disturbed urban watersheds, wastewater treatment facilities play an important role in modulating sediment hysteresis and transport. Wastewater discharges augment connectivity of stream material by providing weather-invariant contribution of “hungry” water (i.e., sediment-starved water with high capacity for erosion) that can transport recently deposited bed sediment and/or erode new bank material.⁵² Spectral analysis indicated anthropogenic activities in urbanized watersheds tend to shift the peak variance of streamflow signals toward the high-frequency region, i.e., disturbances to streamflow become more common. These results suggest that wastewater treatment facility discharge substantially shifts the frequency of maximum streamflow variance from every 6.3 h in the watershed without a facility (Blue River) to every 3.8 h in the watershed with the largest discharge permit (Indian Creek) (Figure 5b). Further,

only Blue River lacked the imprint of several local maxima variance peaks in the more-frequent disturbance periods, corresponding to every 1.5–2.5 h. This is potentially a result of facility discharges that occur at sub-daily frequencies, giving large variations in streamflow values; however, the authors do not have access to sub-daily effluent data to investigate this fully. The scaling exponent (β) in the PSD plots of streamflow (on average ~ 3.94) and turbidity (on average ~ 2.12) (Table S4) were larger than the values reported in similar studies^{23,25} suggesting higher persistence in processes affecting these parameters in our study region. Interestingly, no trends were observed in turbidity variance-peak signals implying that other underlying processes in turbidity dynamics mask the effects of wastewater treatment facility discharges.

The degree and type of anthropogenic disturbance also changed the drivers of sediment transport in each hysteresis category. Large baseflow, for example, promoted counter-clockwise (distal) behavior in highly disturbed rural (Blue River) and urban (Indian Creek) watersheds but clockwise (proximal) behavior in low-disturbance rural watersheds. Larger baseflow is generally indicative of higher hydrological connectivity to streams and therefore this contrasting behavior suggests difference in connectivity extent of distal sediment sources.⁵³ In watersheds with a high degree of disturbance, baseflow preconnects distal pathways to the main channel, either via roadway and/or tile drainage, allowing for continuous delivery of sediment after the initial sediment flush.⁵⁴ For low-disturbance rural watersheds, our results suggest that baseflow wets near-channel material for quick mobilization but distally located material, which is also wetted, encounters arduous disconnectivity along its longitudinal path to the observation point.^{11,55} Larger proportion of counter-clockwise events in urban watersheds can also be attributed to faster delivery of runoff to streams due to the presence of storm drains. This rapid transport of relatively sediment-free water causes peak flow to occur sooner than in rural watersheds⁴⁸ as runoff cannot erode stream channel material fast enough to meet its demands by the time it passes the sensor station. Therefore, although urban watersheds were characterized by rapid sediment transport (i.e., the largest flushing indices; Table S2), turbidity peaks generally appeared after streamflow once sediment from distal tributaries, drainage channels, and roadside ditches could arrive.

Hence, in urban watersheds, distal sediment most likely refers to a higher connectivity of bank material far from the monitoring station rather than upland sediment. A higher percentage of impervious surfaces in urban watersheds results in faster channel flow that can overcome the bed and bank critical shear stress and transport sediment to the most downstream point of the watershed.⁴⁷ Although near-channel sediment sources far from the pour point are considered as key drivers of counter-clockwise hysteresis patterns, well-connected urban hillslope sediment can also result in delayed turbidity peaks during storm events. Recent studies have shown that hillslope sediment coming from construction sites, grass/mulch surfaces, and gravel surfaces play a more important role in sediment transport to stream corridors than previously thought.⁴⁶ Moreover, we observed greater percentage of counter-clockwise events in Mill Creek although this watershed is less urbanized compared to Indian Creek. This is consistent with recent findings that hillslope sediment supply increases with urbanization up to a point, and then decreases again as sediment sources become scarce.⁴⁶ Thus, watershed managers

and planning agencies must take great care to reduce sediment loss, which can be accelerated by urbanization. For example, our results indicate that urbanization augments connectivity of distal sediment; thus, watershed managers should consider best management practices, such as buffer strips and bank stabilization, that not only target zones proximal to the main channel but also those further from it that are well connected.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.1c00740>.

Summary of runoff, antecedent, and rainfall variables (Table S1); percentage of storm events categorized by hysteresis pattern (Table S2); mean event-scale statistics for the studied watersheds (Table S3); spectral slope of 15 min streamflow, turbidity, sediment concentration, and sediment load time series (Table S4); mean daily stream discharge and turbidity sensor data (Figure S1); turbidity sensor regressions against laboratory-measured suspended sediment concentration (Figure S2); sediment flux over the length of study period as a function of hysteresis type for the five studied watersheds (Figure S3); and seasonality of different hysteresis types for each study watershed (Figure S4) (PDF)

Database of all 849 storm events with calculated values for the runoff, antecedent, and rainfall variables for each event (XLSX)

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Notes

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